

Summary of Lecture 1 – INTRODUCTION TO PHYSICS

1. Physics is a science. Science works according to the *scientific method*. The *scientific method* accepts only reason, logic, and experimental evidence to tell between what is scientifically correct and what is not. Scientists do not simply believe – they test, and keep testing until satisfied. Just because some “big scientist” says something is right, that thing does not become a fact of science. Unless a discovery is repeatedly established in different laboratories at different times by different people, or the same theoretical result is derived by clear use of established rules, we do not accept it as a scientific discovery. The real strength of science lies in the fact that it continually keeps challenging itself.
2. It is thought that the laws of physics do not change from place to place. This is why experiments carried out in different countries by different scientists – of any religion or race – have always led to the same results if the experiments have been done honestly and correctly. We also think that the laws of physics today are the same as they were in the past. Evidence, contained in the light that left distant stars billions of years ago, strongly indicates that the laws operating at that time were no different than those today. The spectra of different elements then and now are impossible to tell apart, even though physicists have looked very carefully.
3. This course will cover the following broad categories:
 - a) *Classical Mechanics*, which deals with the motion of bodies under the action of forces. This is often called *Newtonian mechanics* as well.
 - b) *Electromagnetism*, whose objective is to study how charges behave under the influence of electric and magnetic fields as well as understand how charges can create these fields.
 - c) *Thermal Physics*, in which one studies the nature of heat and the changes that the addition of heat brings about in matter.
 - d) *Quantum Mechanics*, which primarily deals with the physics of small objects such as atoms, nuclei, quarks, etc. However, Quantum Mechanics will be treated only briefly for lack of time.
4. Every physical quantity can be expressed in terms of three fundamental dimensions: Mass (M), Length (L), Time (T). Some examples:

$$LT^{-1} \leftarrow \frac{L}{T} \leftarrow \frac{s}{s} \leftarrow$$

Speed	LT^{-1}
Acceleration	LT^{-2}
Force	MLT^{-2}
Energy	ML^2T^{-2}
Pressure	$ML^{-1}T^{-2}$

You cannot add quantities that have different dimensions. So force can be added to force, but force can never be added to energy, etc. A formula is definitely wrong if the dimensions on the left and right sides of the equal sign are different.

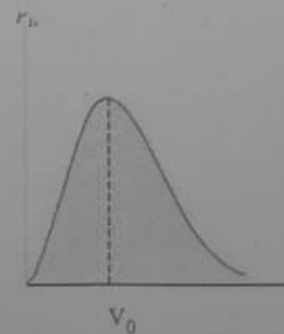
5. Remember that any function $f(x)$ takes as input a dimensionless number x and outputs a quantity f (which may, or may not have a dimension). Take, for example, the function $f(\theta) = \sin \theta$. You know the expansion: $\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$. If θ had a dimension then you would be adding up quantities of different dimensions, and that is not allowed.

6. Do not confuse units and dimensions. We can use different units to measure the same physical quantity. So, for example, you can measure the mass in units of kilograms, pounds, or even in *sair* and *chatak*! In this course we shall always use the MKS or Metre-Kilogram-Second system. When you want to convert from one system to another, be methodical as in the example below:

$$1 \frac{\text{mi}}{\text{hr}} = 1 \frac{\text{mi}}{\text{hr}} \times 5280 \frac{\text{ft}}{\text{mi}} \times \frac{1}{3.28} \frac{\text{m}}{\text{ft}} \times \frac{1}{3600} \frac{\text{hr}}{\text{s}} = 0.447 \frac{\text{m}}{\text{s}}$$

When you write it out in this manner, note that various quantities cancel out cleanly in the numerator and denominator. So you never make a mistake!

7. A good scientist first thinks of the larger picture and then of the finer details. So, estimating orders of magnitude is extremely important. Students often make the mistake of trying to get the decimal points right instead of the first digit – which obviously matters the most! So if you are asked to calculate the height of some building using some data and you come up with 0.301219 metres or 4.01219×10^6 metres, then the answer is plain nonsense even though you may have miraculously got the last six digits right. Physics is commonsense first, so use your intelligence before submitting any answer.
8. Always check your equations to see if they have the same dimensions on the left side as on the right. So, for example, from this principle we can see the equation $v^2 = u^2 + 2at$ is clearly wrong, whereas $v^2 = u^2 + 13a^2t^2$ could possibly be a correct relation. (Here v and u are velocities, a is acceleration, and t is time.) Note here that I use the word *possibly* because the dimensions on both sides match up in this case.
9. Whenever you derive an equation that is a little complicated, see if you can find a special limit where it becomes simple and transparent. So, sometimes it is helpful to imagine that some quantity in it is very large or very small. Where possible, make a “mental graph” so that you can picture an equation. So, for example, a formula for the distribution of molecular speeds in a gas could look like $f(v) = ve^{-(v-v_0)^2/a^2}$. Even without knowing the value of a you can immediately see that
- $f(v)$ goes to zero for large values of v , and $v = 0$.
 - The maximum value of $f(v)$ occurs at v_0 and the function decreases on both side of this value.



Summary of Lecture 2 - KINEMATICS I

shortest distance by the time

1. $x(t)$ is called displacement and it denotes the position of a body at time. If the displacement is positive then that body is to the right of the chosen origin and if negative, then it is to the left. Displacement $v = v_2 - v_1$

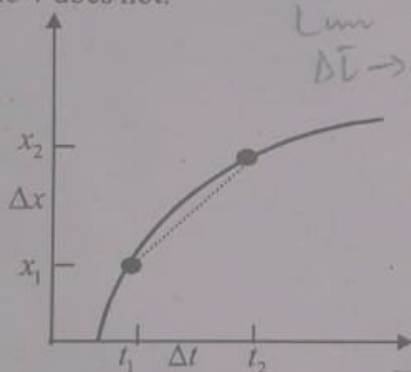
2. If a body is moving with average speed v then in time t it will cover a distance $d = vt$. But, in fact, the speed of a car changes from time to time and so one should limit the use of this formula to small time differences only. So, more accurately, one defines an average speed over the small time interval Δt as:

$$v = \frac{x_2 - x_1}{\Delta t} \quad \text{average speed} = \frac{\text{distance travelled in time } \Delta t}{\Delta t}$$

3. We define instantaneous velocity at any time t as:

$$v = \frac{x(t_2) - x(t_1)}{t_2 - t_1} \equiv \frac{\Delta x}{\Delta t}$$

Here Δx and Δt are both very small quantities that tend to zero but their ratio v does not.



4. Just as we have defined velocity as the rate of change of distance, similarly we can define instantaneous acceleration at any time t as:

$$a = \frac{v(t_2) - v(t_1)}{t_2 - t_1} \equiv \frac{\Delta v}{\Delta t}$$

Here Δv and Δt are both very small quantities that tend to zero but their ratio a is not zero, in general. Negative acceleration is called deceleration. The speed of a decelerating body decreases with time.

5. Some students are puzzled by the fact that a body can have a very large acceleration but can be standing still at a given time. In fact, it can be moving in the opposite direction to its acceleration. There is actually nothing strange here because position,

$$t = \frac{v - v_0}{a} \Rightarrow x = x_0 + v_0 \left(\frac{v - v_0}{a} \right) + \frac{1}{2} a \left(\frac{v - v_0}{a} \right)^2 \Rightarrow x - x_0 = \frac{v^2 - v_0^2}{2a} \Rightarrow v^2 = v_0^2 + 2a(x - x_0)$$

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$v = at + v_0$
constant speed $\rightarrow x = x_0 + vt$

6. For constant acceleration and a body that starts from rest at $t = 0$, v increases linearly with time, $v \propto t$ (or $v = at$). If the body has speed v_0 at $t = 0$, then at time t , $v = at + v_0$. $\rightarrow$ constant velocity

$$\frac{x - x_0}{t - 0} = \frac{v + v_0}{2}$$

$$\Rightarrow \frac{v_0 + at + v_0}{2}$$

$$\Rightarrow \frac{2v_0 + at}{2}$$

$$\Rightarrow v_0 + \frac{1}{2}at$$

$$x - x_0 = v_0 t + \frac{1}{2}at^2$$

$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

7. We know in (6) above how far a body moving at constant speed moves in time t . But what if the body is changing its speed? If the speed is increasing linearly (i.e. constant acceleration), then the answer is particularly simple: just use the same formula as in (6) but use the average speed: $(v_0 + v_0 + at)/2$. So we get

$x = x_0 + (v_0 + v_0 + at)t/2 = x_0 + v_0 t + \frac{1}{2}at^2$. This formula tells you how far a body moves in time t if it moves with constant acceleration a , and if started at position x_0 at $t=0$ with speed v_0 .

- We can eliminate the time using (7), and arrive at another useful formula that tells us what the final speed will be after the body has traveled a distance equal to

$x - x_0$ after time t , $v^2 = v_0^2 + 2a(x - x_0)$.

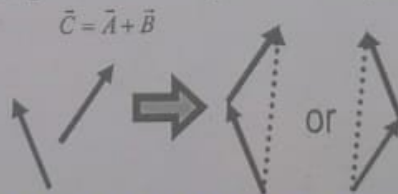
9. Vectors: a quantity that has a size as well as direction is called a **vector**. So, for example, the wind blows with some speed and in some direction. So the **wind velocity** is a vector.

10. If we choose axes, then a vector is fixed by its components along those axes. In one dimension, a vector has only one component (call it the x-component). In two dimensions, a vector has both x and y components. In three dimensions, the components are along the x, y, z axes.

11. If we denote a vector $\vec{r} = (x, y)$ then, $r_x = x = r \cos \theta$, and $r_y = y = r \sin \theta$.

Note that $x^2 + y^2 = r^2$. Also, that $\tan \theta = y/x$.

13. Two vectors can be added together geometrically. We take any one vector, move it without changing its direction so that both vectors start from the same point, and then make a parallelogram. The diagonal of the parallelogram is the resultant.



14. Two vectors can also be added algebraically. In this case, we simply add the components of the two vectors along each axis separately. So, for example, Two vectors can be put together as $(1.5, 2.4) + (1, -1) = (2.5, 1.4)$.

$$C = \sqrt{C_x^2 + C_y^2}$$

$$\theta = \arctan\left(\frac{C_y}{C_x}\right) \text{ or } \tan^{-1}\left(\frac{C_y}{C_x}\right)$$

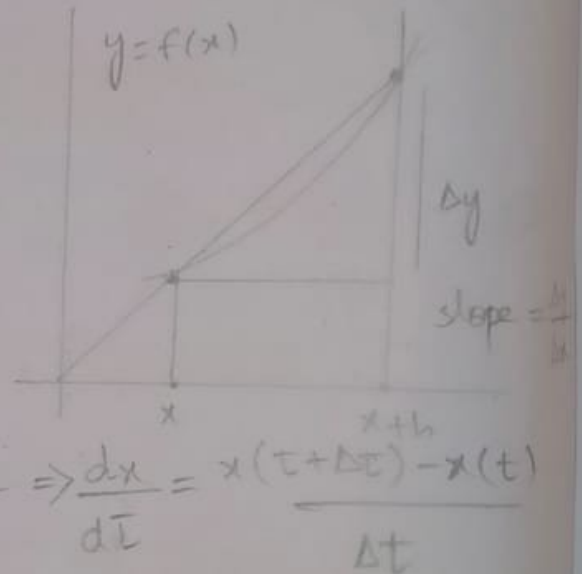
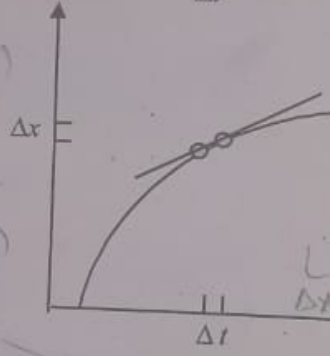
Summary of Lecture 3 – KINEMATICS II

1. The concept of the derivative of a function is exceedingly important. The derivative shows how fast a function changes when its argument is changed. (Remember that for $f(x)$ we say that f is a function that depends upon the argument x . You should think of f as a machine that gives you the value f when you input x .)
2. Functions do not always have to be written as $f(x)$. $x(t)$ is also a function. It tells us where a body is at different times t .

3. The derivative of $x(t)$ at time t is defined as:

$$\frac{dx}{dt} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$$

$$= \lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}$$



4. Let's see how to calculate the derivative of a simple function like $x(t) = t^2$. We must first calculate the difference in x at two slightly different values, t and $t + \Delta t$, while remembering that we choose Δt to be extremely small:

$$\Delta x = (t + \Delta t)^2 - t^2$$

$$= t^2 + (\Delta t)^2 + 2t\Delta t - t^2$$

$$\frac{\Delta x}{\Delta t} = \Delta t + 2t \Rightarrow \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} = 2$$

5. In exactly the same way you can show that if $x(t) = t^n$ then:

$$\frac{dx}{dt} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = nt^{n-1}$$

This is an extremely useful result.

6. Let us apply the above to the function $x(t)$ which represents the distance moved by a body with constant acceleration (see lecture 2):

$$x(t) = x_0 + v_0 t + \frac{1}{2} a t^2$$

$$\frac{dx}{dt} = 0 + v_0 + \frac{1}{2} a (2t) = v_0 + a t$$

This clearly shows that $\frac{dv}{dt} = 0 + a = a$ (acceleration is constant)

7. A stone dropped from rest increases its speed in the downward direction according to $\frac{dv}{dt} = g \approx 9.8 \text{ m/sec}$. This is true provided we are fairly close to the earth, otherwise the value of g decreases as we go further away from the earth. Also, note that if we measured distances from the ground up, then the acceleration would be negative.

↑ -ve decrease
↓ increase +ve

8. A useful notation: write $\frac{dv}{dt} = \frac{d}{dt} \left(\frac{dx}{dt} \right) = \frac{d^2 x}{dt^2}$. We call $\frac{d^2 x}{dt^2}$ the second derivative of x with respect to t , or the rate of rate of change of x with respect to t .

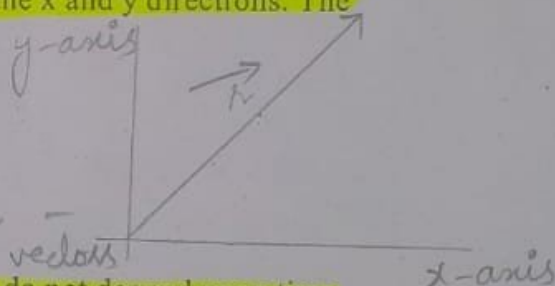
9. It is easy to extend these ideas to a body moving in both the x and y directions. The position and velocity in 2 dimensions are:

$$s = vt \\ d = vt \\ \Rightarrow v = \frac{d}{t}$$

$$\vec{r} = x(t)\hat{i} + y(t)\hat{j}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$$

$$= v_x\hat{i} + v_y\hat{j} \rightarrow \text{along with unit vectors}$$



Here the unit vectors $\hat{i}$ and $\hat{j}$ are fixed, meaning that they do not depend upon time.

10. The scalar product of two vectors $\vec{A}$ and $\vec{B}$ is defined as:

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

You can think of:

$$\vec{A} \cdot \vec{B} = (A)(B \cos \theta)$$

$$= (\text{length of } \vec{A}) \times (\text{projection of } \vec{B} \text{ on } \vec{A})$$

OR,

$$\vec{A} \cdot \vec{B} = (B)(A \cos \theta)$$

$$= (\text{length of } \vec{B}) \times (\text{projection of } \vec{A} \text{ on } \vec{B}).$$

Remember that for unit vectors $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = 1$ and $\hat{i} \cdot \hat{j} = 0$.

self product $\rightarrow 1$

Object $\rightarrow$ Anything that has mass and occupies space is called object/matter.

Summary of Lecture 4 - FORCE AND NEWTON'S LAWS

Force $\rightarrow$

1. Ancient view: objects tend to stop if they are in motion; force is required to keep something moving. This was a natural thing to believe in because we see objects stop moving after some time; frictionless motion is possible to see only in rather special circumstances.
2. Modern view: objects tend to remain in their initial state; force is required to change motion. Resistance to changes in motion is called *inertia*. More inertia means it is harder to make a body accelerate or decelerate.
3. Newton's First Law: An object will remain at rest or move with constant velocity unless acted upon by a net external force. (A non-accelerating reference frame is called an *inertial* frame; Newton's First Law holds only in inertial frames.)
4. More force leads to more acceleration: $\Rightarrow a \propto F$
5. The greater the mass of a body, the harder it is to change its state of motion. More mass means more inertia. In other words, more mass leads to less acceleration:

$$\Rightarrow a \propto \frac{1}{m}$$

Combine both the above observations to conclude that:

$$a \propto \frac{F}{m} \Rightarrow a = k \frac{F}{m} \quad \because k=1 \quad \text{so} \quad a = \frac{F}{m}$$

6. Newton's Second Law: $a = \frac{F}{m}$ (or, if you prefer, write as $F = ma$).
7. $F = ma$ is one relation between three independent quantities (m, a, F). For it to be useful, we must have separate ways of measuring mass, acceleration, and force. Acceleration is measured from observing the rate of change of velocity; mass is a measure of the amount of matter in a body (e.g. two identical cars have twice the mass of a single one). Forces (due to gravity, a stretched spring, repulsion of two like charges, etc) will be discussed later.

$$m = M, a = LT^{-2} \Rightarrow F = ma = MLT^{-2}$$

8. Force has dimensions of $[\text{mass}] \times [\text{acceleration}] = MLT^{-2}$. In the MKS system the unit of force is the Newton. It has the symbol N where:

$$1 \text{ Newton} = 1 \text{ kilogram.metre/second}^2. \quad 1N = 1kgms^{-2}$$

9. Forces can be internal or external. For example the mutual attraction of atoms within a block of wood are called internal forces. Something pushing the wood

is an external force. In the application of $F = ma$, remember that F stands for the total external force upon the body.

10. Forces are vectors, and so they must be added vectorially:

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots$$

This means that the components in the $\hat{x}$ direction must be added separately, those in the $\hat{y}$ direction separately, etc.

11. Gravity acts directly on the mass of a body - this is a very important experimental observation due to Newton and does not follow from $F = ma$. So a body of mass m_1 experiences a force $F_1 = m_1g$ while a body of mass m_2 experiences a force $F_2 = m_2g$, where g is the acceleration with which any body (big or small) falls under the influence of gravity. (Galileo had established this important fact when he dropped different masses from the famous leaning tower of Pisa!)

12. The weight of a body W is the force which gravity exerts upon it, $W = mg$. Mass and weight are two completely different quantities. So, for example, if you used a spring balance to weigh a kilo of grapes on earth, the same grapes would weigh only 1/7 kilo on the moon.

13. Newton's Third Law: for every action there is an equal and opposite reaction. More precisely, $F_{AB} = -F_{BA}$, where F_{AB} is the force exerted by body B upon A whereas F_{BA} is the force exerted by body A upon B . Ask yourself what would happen if this was not true. In that case, a system of two bodies, even if it is completely isolated from the surroundings, would have a net force acting upon it because the net force acting upon both bodies would be $F_{AB} + F_{BA} \neq 0$. → free fall

14. If action and reaction are always equal, then why does a body accelerate at all? Students are often confused by this. The answer: in considering the acceleration of a body you must consider only the (net) force acting upon that body. So, for example, the earth pulls a stone towards it and causes it to accelerate because there is a net force acting upon the stone. On the other hand, by the Third Law, the stone also pulls the earth towards it and this causes the earth to accelerate towards the stone. However, because the mass of the earth is so large, we are only able to see the acceleration of the stone and not that of the earth.

Equilibrium $\rightarrow$ A body is in equilibrium when all the forces acting upon it must be zero. VU

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Summary of Lecture 5 - APPLICATIONS OF NEWTON'S LAWS - I

1. An obvious conclusion from $F = ma$ is that if $F = 0$ then $a = 0$! How simple, yet how powerful! This says that for any body that is not accelerating the sum of all the forces acting upon it must vanish.

$$\sum \vec{F} = F_x + F_y = 0$$

2. Examples of systems in equilibrium: a stone resting on the ground; a pencil balanced on your finger; a ladder placed against the wall, an aircraft flying at a constant speed and constant height.
3. Examples of systems out of equilibrium: a stone thrown upwards that is at its highest point; a plane diving downwards; a car at rest whose driver has just stepped on the car's accelerator.

4. If you know the acceleration of a body, it is easy to find the force that causes it to accelerate. Example: An aircraft of mass m has position vector,

$$\vec{r} = (at + bt^3)\hat{i} + (ct^2 + dt^4)\hat{j}$$

What force is acting upon it?

SOLUTION:

$$\vec{F} = m \frac{d^2x}{dt^2} \hat{i} + m \frac{d^2y}{dt^2} \hat{j}$$

$$= 6bmt\hat{i} + m(2c + 12dt^2)\hat{j}$$

$$\vec{r} = r_x \hat{i} + r_y \hat{j}$$

$$\vec{r} = (x)\hat{i} + (y)\hat{j}$$

5. The other way around is not so simple: suppose that you know F and you want to find x . For this you must solve the equation,

$$\frac{d^2x}{dt^2} = \frac{F}{m}$$

This may or may not be easy, depending upon F (which may depend upon both x as well as t if the force is not constant).

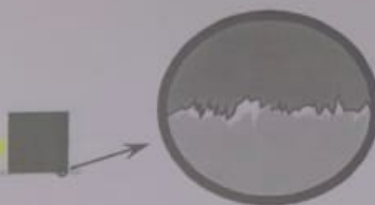
$$\sum F = T_2 - T_1 = ma \text{ if } m=0 \text{ then } T_2 = T_1$$

6. Ropes are useful because you can pull from a distance to change the direction of a force. The tension, often denoted by T , is the force you would feel if you cut the rope and grabbed the ends. For a massless rope (which may be a very good approximation in many situations) the tension is the same at every point along the rope. Why? Because if you take any small slice of the rope it weighs nothing (or very little). So if the force on one side of the slice was any different from the force on the other side, it would be accelerating hugely. All this was for the "ideal rope" which has no mass and never breaks. But this idealization is often good enough.

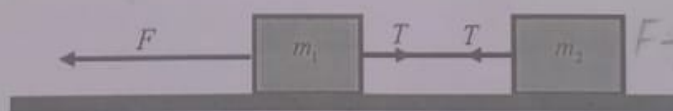
7. We are all familiar with frictional force. When two bodies rub against each other, the frictional force acts upon each body separately opposite to its direction of motion (i.e. it acts to slow down the motion). The harder you press two bodies against each other, the greater the friction. Mathematically, $\vec{F} = \mu \vec{N}$, where $\vec{N}$ is the force with which you press the two bodies against each other (normal force). The quantity μ is called the coefficient of friction (obviously!). It is large for rough surfaces, and small for smooth ones. Remember that $\vec{F} = \mu \vec{N}$ is an empirical relation and holds only approximately. This is obviously true: if you put a large enough mass on a table, the table will start to bend and will eventually break.

8. Friction is caused by roughness at a microscopic level

- if you look at any surface with a powerful microscope you will see unevenness and jaggedness. If these big bumps are levelled somehow, friction will still not disappear because there will still be little bumps due to atoms. More precisely, atoms from the two bodies will interact each other because of the electrostatic interaction between their charges. Even if an atom is neutral, it can still exchange electrons and there will be a force because of surrounding atoms.



9. Consider the two blocks below on a frictionless surface:



$$T = \frac{m_2 F}{m_1 + m_2}$$

$$F - T = m_1 a \quad \text{--- (1)}$$

$$T = m_2 a \quad \text{--- (2)}$$

$$F - T + T = m_1 a + m_2 a$$

$$F = a(m_1 + m_2)$$

$$a = \frac{F}{m_1 + m_2}$$

We want to find the tension and acceleration: The total force on the first mass is $F - T$ and so $F - T = m_1 a$. The force on the second mass is simply T and so $T = m_2 a$. Solving the

above, we get: $T = \frac{m_2 F}{m_1 + m_2}$ and $a = \frac{F}{m_1 + m_2}$.

10. There is a general principle by which you solve equilibrium problems. For equilibrium, the sum of forces in every direction must vanish. So $F_x = F_y = F_z = 0$. You may always choose the x , y , z directions according to your convenience. So, for example, as in the lecture problem dealing with a body sliding down an inclined plane, you can choose the directions to be along and perpendicular to the surface of the plane.

$$\text{Normal force} = -\text{Fluid force}$$

$$F = -F$$

Summary of Lecture 6 – APPLICATIONS OF NEWTON'S LAWS – II

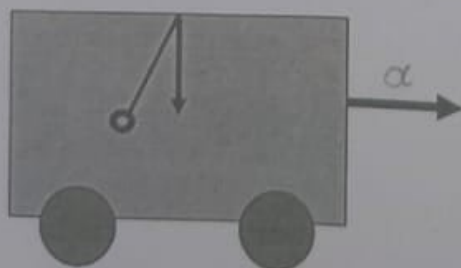
1. As a body moves through a fluid it displaces the fluid. It has to exert a force on the fluid to push it out of the way. By Newton's third law, the fluid pushes back on the body with an equal and opposite force. The direction of the fluid resistance force on a body is always opposite to the direction of the body's velocity relative to the fluid.

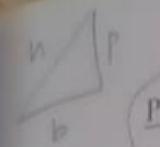
2. The magnitude of the fluid resistance force usually increases with the speed of the body through the fluid. Typically, $f = kv$ (an empirical law!). Imagine that you drop a ball bearing into a deep container filled with oil. After a while the ball bearing will approach its maximum (terminal) speed when the forces of gravity and friction balance each other: $mg = kv$ from which $v_{\text{final}} = mg/k$.

3. The above was a simple example of equilibrium under two forces. In general, while solving problems you should a) draw a diagram, b) define an origin for a system of coordinates, c) identify all forces (tension, normal, friction, weight, etc) and their x and y components, d) Apply Newton's law separately along the x and y axes. e) find the accelerations, then velocities, then displacements. This sounds very cook-book, and in fact it will occur to you naturally how to do this while solving actual problems.

4. Your weight in a lift: suppose you are in a lift that is at rest or moving at constant velocity. In either case $a=0$ and the normal force N and the force due to gravity are exactly equal, $N - Mg = 0 \Rightarrow N = Mg$. But if the lift is accelerating downwards then $Mg - N = Ma$ or $N = M(g - a)$. So now the normal force (i.e. the force with which the floor of the lift is pushing on you) is decreased. Note that if the lift is accelerating downwards with acceleration a (which it will if the cable breaks!) then $N=0$ and you will experience weightlessness just like astronauts in space do. Finally, if the lift is accelerating upwards then a is negative and you will feel heavier.

5. Imagine that you are in a railway wagon and want to know how much you are accelerating. You are not able to look out of the windows. A mass is hung from the roof. Find the acceleration of the car from the angle made by the mass.





$$T \sin \theta = ma$$

$$T \cos \theta = mg$$

$$\therefore \tan \theta = \frac{\sin}{\cos} \Rightarrow \frac{T \sin \theta}{T \cos \theta} = \frac{ma}{mg} \Rightarrow \tan \theta = \frac{a}{g}$$

$$\sin = \frac{p}{h}$$

We first balance the forces vertically: $T \cos \theta = mg$

And then horizontally: $T \sin \theta = ma$

$$\cos = \frac{b}{h}$$

From these two equations we find that: $\tan \theta = \frac{a}{g}$

Note that the mass m doesn't matter - it cancels out!



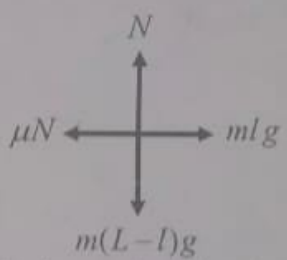
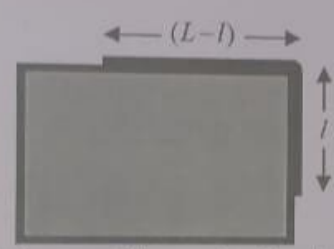
6. Friction is a funny kind of force. It does not make up its mind which way to act until some other force compels it to decide. Imagine a block lying on the floor. If you push it forward, friction will act backward. And if you push it to the left, friction will act to the right. In other words, the direction of the frictional force is always in the opposite direction to the applied force.

7. Let us solve the following problem: a rope of total length L and mass per unit length m is put on a table with a length l hanging from one edge. What should be l such that the rope just begins to slip?

$$N = m(L-l)g \quad \text{--- (1)}$$

$$\mu N = mlg \quad \text{--- (2)}$$

$$N = \frac{mlg}{\mu}$$

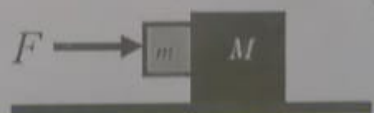


Sub (1) & (2) To solve this, look at the balance of forces in the diagram below: in the vertical direction, the normal force balances the weight of that part of the rope that lies on the table:

$0 = mL - mlg$ $(N = m(L-l)g)$. In the horizontal direction, the rope exerts a force mlg to the right, which is counteracted by the friction that acts to the left. Therefore $\mu N = mlg$. Substituting N

from the first equation we find that $l = \frac{\mu L}{\mu + 1}$. Note that if μ is very small then even a small piece of string that hangs over the edge will cause the entire string to slip down.

9. In this problem, we would like to calculate the minimum force F such that the small block does not slip downwards. Clearly, since the 2 bodies move together, $F = (m+M)a$. This gives



$a = \frac{F}{(m+M)}$. We want the friction μN to be at least as large as the downwards force, mg .

So, we put $N = ma = m \left(\frac{F}{(m+M)} \right)$ from which the minimum horizontal force needed to

prevent slippage is $F = \frac{(m+M)g}{\mu}$

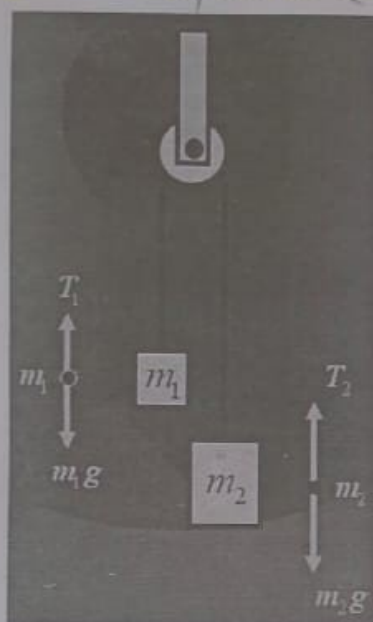
$$\mu \left[\frac{\mu + 1}{\mu} \right] l g = m l g$$

$$l = \frac{\mu L}{\mu + 1}$$

PHYSICS - PHY101

There is an error in video lecture # 6 (video recording at 41:00) please consider the following problem with correct solution.

Consider two bodies of unequal masses m_1 and m_2 connected by the ends of a string, which passes over a frictionless pulley as shown in the diagram.



upward $\leftarrow F_1 = T - m_1g$ $\because W = m_1g$
 $F = T - W_1$
 $m_1a = T - m_1g \text{ --- (1)}$

$F_2 = W_2 - T$
 $m_2a = m_2g - T \text{ --- (2)}$

$m_2 > m_1$

$T - m_1g = m_1a$

$m_2g - T = m_2a$

$m_2g - m_1g = (m_1 + m_2)a$

For acceleration $\rightarrow a = \frac{(m_2 - m_1)}{(m_1 + m_2)}g$

$m_2 > m_1$

$T - m_1g = m_1a$

$a = \frac{T - m_1g}{m_1}$

$m_2g - T = m_2a$

$a = \frac{m_2g - T}{m_2}$

equating (i) and (ii)

cross multiplication $\leftarrow \frac{T - m_1g}{m_1} = \frac{m_2g - T}{m_2}$

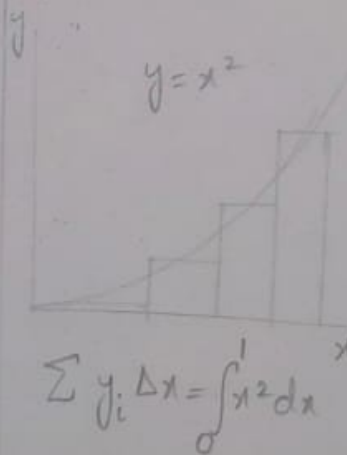
$m_2T - m_1m_2g = m_1m_2g - m_1T$

$m_2T + m_1T = m_1m_2g + m_1m_2g$

$T(m_2 + m_1) = 2m_1m_2g$

For Tension $\rightarrow T = \frac{2m_1m_2g}{(m_2 + m_1)}$

lecture # 07 (Example)



No. of rectangles: $n=4$

$\Delta x = \frac{1-0}{4} = 0.25$

$x_i = (i - 1/2)\Delta x$

$= (i - 1/2)/n$

$y_i = x_i^2 = (i - 1/2)^2/n^2$

$\sum y_i \Delta x = \int_0^1 x^2 dx$

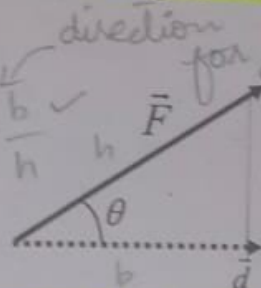
$\sum_{n=1}^i y_i \Delta x = \sum_{n=1}^i \frac{(i - 1/2)^2}{n^2}$

$= \frac{1}{3} - \frac{1}{12n^2} \Rightarrow 0.328$

Summary of Lecture 7 - WORK AND ENERGY

1. Definition of work: force applied in direction of displacement $\times$ displacement. This means that if the force F acts at an angle θ with respect to the direction of motion, then

$$W = \vec{F} \cdot \vec{d} = Fd \cos \theta$$



When F and d are $\perp$ then no work is done.

Positive work: $\theta = 0^\circ$

Negative work: $\theta = 180^\circ$

Zero work: $\theta = 90^\circ$

2. a) Work is a scalar - it has magnitude but no direction.

b) Work has dimensions: $M \times (L T^{-2}) \times L = M L^2 T^{-2}$

c) Work has units: 1 Newton $\times$ 1 Metre $\equiv$ 1 Joule (J) $\Rightarrow Nm = J$

3. Suppose you lift a mass of 20 kg through a distance of 2 metres. Then the work you do is 20 kg $\times$ 9.8 Newtons $\times$ 2 metres = 392 Joules. On the other hand, the force of gravity is directed opposite to the force you exert and the work done by gravity is -392 Joules.

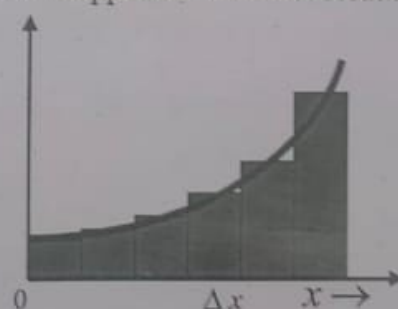
4. What if the force varies with distance (say, a spring pulls harder as it becomes longer). In that case, we should break up the distance over which the force acts into small pieces so that the force is approximately constant over each bit. As we make the pieces smaller and smaller, we will approach the exact result:

$$\Delta W_1 = F_1 \Delta x$$

$$\Delta W_2 = F_2 \Delta x$$

$$\Delta W_3 = F_3 \Delta x$$

$$\Delta W_n = F_n \Delta x$$



The total work done by F in moving a body from x_i to x_f is:-

$$W = \int_{x_i}^{x_f} F(x) dx$$

↑ variable force

Now add up all the little pieces of work:

$$W = \Delta W_1 + \Delta W_2 + \dots + \Delta W_N = F_1 \Delta x + F_2 \Delta x + \dots + F_N \Delta x \equiv \sum_{n=1}^N F_n \Delta x$$

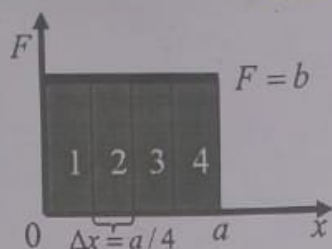
To get the exact result let $\Delta x \rightarrow 0$ and the number of intervals $N \rightarrow \infty$: $W = \lim_{\Delta x \rightarrow 0} \sum_{n=1}^N F_n \Delta x$

Definition: $W = \lim_{\Delta x \rightarrow 0} \sum_{n=1}^N F_n \Delta x \equiv \int_{x_i}^{x_f} F(x) dx$ is called the integral of F with respect to x from

x_i to x_f . This quantity is the work done by a force, constant or non-constant. So if the force is known as a function of position, we can always find the work done by calculating the definite integral.

5. Just to check what our result looks like for a constant force, let us calculate W if $F = b$,

constant force



$$\frac{1}{4}a(b) + \frac{1}{4}a(b) + \frac{1}{4}a(b) + \frac{1}{4}a(b) = ab$$

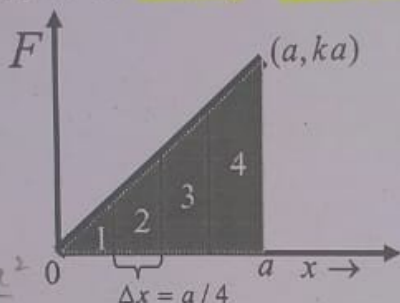
$$\int_0^a F dx = b \times c$$

$$\int_0^a F dx = \int_0^a b dx$$

$$\therefore \int_0^a F dx = ab \Rightarrow b \int_0^a dx \Rightarrow b[x]_0^a$$

6. Now for a less trivial case: suppose that $F = kx$, i.e. the force increases linearly with x .

variable force



$$\int_0^a F dx = \int_0^a kx dx$$

$$\Rightarrow k \int_0^a x dx$$

$$\Rightarrow k \left[\frac{x^2}{2} \right]_0^a \Rightarrow k \times \frac{a^2}{2}$$

$$\Rightarrow b(a-0)$$

$$\Rightarrow b \times a$$

$$W = b \times a \text{ or } ab$$

$$\text{Area of shaded region} = \frac{1}{2}(a)(ka) = k \frac{a^2}{2}$$

$$\therefore \int_0^a F dx = k \frac{a^2}{2}$$

7. Energy is the capacity of a physical system to do work:

- it comes in many forms – mechanical, electrical, chemical, nuclear, etc
- it can be stored
- it can be converted into different forms
- it can never be created or destroyed

8. Accepting the fact that energy is conserved, let us derive an expression for the kinetic energy of a body. Suppose a constant force accelerates a mass m from speed 0 to speed v over a distance d . What is the work done by the force? Obviously the answer is:

$$W = Fd. \text{ But } F = ma \text{ and } v^2 = 2ad. \text{ This gives } W = (ma)d = \frac{mv^2}{2d} d = \frac{1}{2}mv^2. \text{ So, we}$$

conclude that the work done by the force has gone into creating kinetic energy. and that the amount of kinetic energy possessed by a body moving with speed v is $\frac{1}{2}mv^2$.

9. The work done by a force is just the force multiplied by the distance – it does not depend upon time. But suppose that the same amount of work is done in half the time. We then say that the power is twice as much.

We define:

$$\text{Power} = \frac{\text{Work done}}{\text{Time taken}}$$

If the force does not depend on time: $\frac{\text{Work}}{\text{Time}} = \frac{F \Delta x}{\Delta t} = F v$. Therefore, $\text{Power} = F v$.

$$1 \text{ hp} = 746 \text{ W} = 0.746 \text{ kW}$$

10. Let's work out an example. A constant force accelerates a bus (mass m) from speed v_1 to speed v_2 over a distance d . What work is done by the engine?

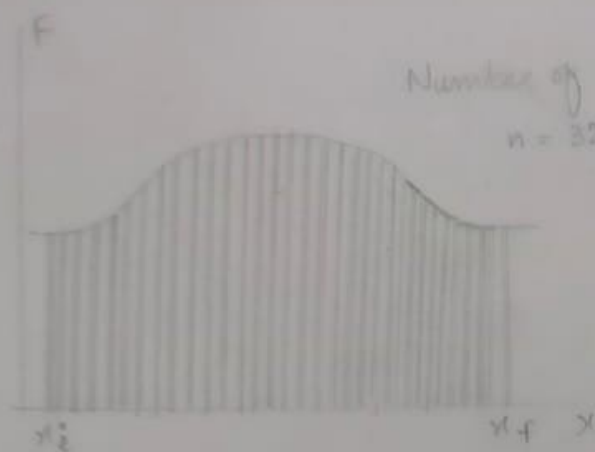


Recall that for constant acceleration, $v_2^2 - v_1^2 = 2a(x_2 - x_1)$ where: v_2 = final velocity,

x_2 = final position, v_1 = initial velocity, x_1 = initial position. Hence, $a = \frac{v_2^2 - v_1^2}{2d}$. Now

calculate the work done: $W = Fd = mad = m \frac{v_2^2 - v_1^2}{2d} d = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$. So the

work done has resulted in an increase in the quantity $\frac{1}{2}mv^2$, which is kinetic energy.



Number of rectangles:
 $n = 32$

Area under the curve b/w x_i and x_f is approximately
 $\sum_{i=1}^n F_i \Delta x$ { if $\Delta x \rightarrow 0$ then the path is smooth }

Summary of Lecture 8 - CONSERVATION OF ENERGY

Energy: The ability of a body to do work. Potential energy is, as the word suggests, the energy "locked up" up somewhere and which can do work. An object can store energy as the result of its position. Stored energy of position is referred to as potential energy. Potential energy is the stored energy of position possessed by an object. It has the ability or capacity to do work like other forms of energies. Potential energy can be converted into kinetic energy, $\frac{1}{2}mv^2$. As I showed you earlier, this follows directly from Newton's Laws.

2. If you lift a stone of mass m from the ground up a distance x , you have to do work against gravity. The (constant) force is mg , and so $W = mgx$. By conservation of energy, the work done by you was transformed into gravitational potential energy whose value is exactly equal to mgx . Where is the energy stored? Answer: it is stored neither in the mass or in the earth - it is stored in the gravitational field of the combined system of stone+earth.

$$F = mg, W = Fd \text{ so } W = mgx$$

3. Suppose you pull on a spring and stretch it by an amount x away from its normal (equilibrium) position. How much energy is stored in the spring? Obviously, the spring gets harder and harder to pull as it becomes longer. When it is extended by length x and you pull it a further distance dx , the small amount of work done is $dW = Fdx = kx dx$. Adding up all the small bits of work gives the total work:

$$W = \int_0^x F dx = \int_0^x kx dx = \frac{1}{2} kx^2$$

$F = kx$
 $dF = k dx$
 $W = \int F dx = \int kx dx$

This is the work you did. Maybe you got tired working so hard. What was the result of your working so hard? Answer: this work was transformed into energy stored in the spring. The spring contains energy exactly equal to $\frac{1}{2}kx^2$.

4. Kinetic energy obviously depends on the frame you choose to measure it in. If you are running with a ball, it has zero kinetic energy with respect to you. But someone who is standing will see that it has kinetic energy! Now consider the following situation: a box of mass 12kg is pushed with a constant force so that its speed goes from zero to 1.5m/sec (as measured by the person at rest on the cart) and it covers a distance of 2.4m. Assume there is no friction.

The spring pushes/pulls with a restoring force proportional to the extension x :
 $F_{\text{spring}} = -kx$

proportionality constant.

work done by external force gives the elastic potential energy:

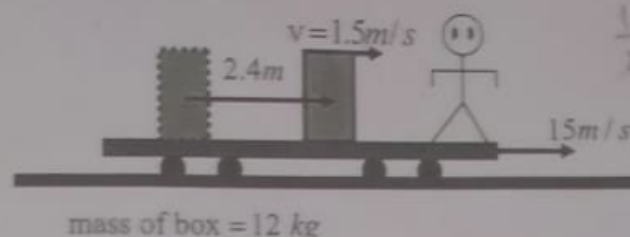
$$W = \int_0^x F dx = \int_0^x kx dx = \frac{1}{2} kx^2$$

8,000 liters of blood is pumped by heart in a day.

work does not depend on path taken. Gravity. Electric. Magnetic.

P.E and K.E will be conserved in the absence of friction.

Example 1: (ball)
 $\frac{1}{2}mv_0^2 = mgh$
 $\Rightarrow h = \frac{v_0^2}{2g}$



Example 2: (spring)

$$\frac{1}{2}mv_0^2 = \frac{1}{2}kx^2$$

$$\Rightarrow \sqrt{v_0^2} = \sqrt{\frac{k}{m}x^2}$$

$$\Rightarrow v_0 = \sqrt{\frac{k}{m}x}$$

Let's first calculate the change in kinetic energy:

$$\Delta K = K_f - K_i = \frac{1}{2}(12\text{kg})(1.5\text{m/s})^2 - 0 = 13.5\text{J}$$

And then the (constant) acceleration:

$$a = \frac{v_f^2 - v_i^2}{2(x_f - x_i)} = \frac{(1.5\text{m/s})^2 - 0}{2(2.4\text{m})} = 0.469\text{m/s}^2$$

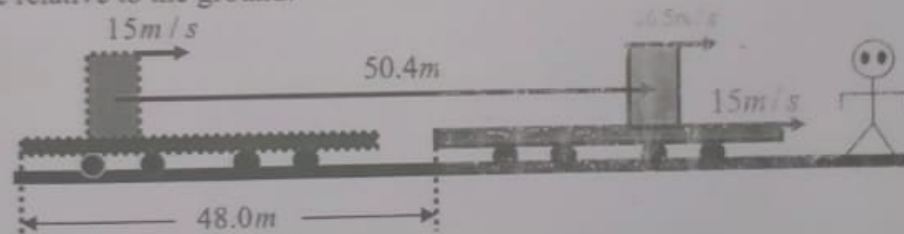
This acceleration results from a constant net force given by:

$$F = ma = (12\text{kg})(0.469\text{m/s}^2) = 5.63\text{N}$$

From this, the work done on the crate is:

$$W = F\Delta x = (5.63\text{N})(2.4\text{m}) = 13.5\text{J} \text{ (same as } \Delta K = 13.5\text{J !)}$$

5. Now suppose there is somebody standing 15 m/sec relative to the ground:



Let us repeat the same calculation:

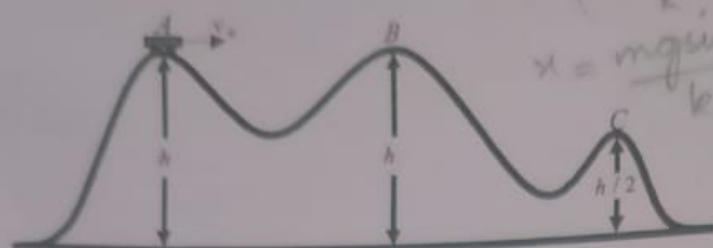
$$\Delta K' = K'_f - K'_i = \frac{1}{2}mv_f'^2 - \frac{1}{2}mv_i'^2$$

$$= \frac{1}{2}(12\text{kg})(16.5\text{m/s})^2 - \frac{1}{2}(12\text{kg})(15.0\text{m/s})^2 = 284\text{J}$$

This example clearly shows that work and energy have different values in different frames.

6. The total mechanical energy is: $E_{\text{mech}} = KE + PE$. If there is no friction then E_{mech} is conserved. This means that the sum does not change with time. For example: a ball is thrown upwards at speed v_0 . How high will it go before it stops? The loss of potential energy is equal to the gain of potential energy. Hence, $\frac{1}{2}mv_0^2 = mgh \Rightarrow h = \frac{v_0^2}{2g}$.

Now look at the smooth, frictionless motion of a car over the hills below:



Even though the motion is complicated, we can use the fact that the total energy is a constant to get the speeds at the points B, C, D:

$$\text{At point A: } \frac{1}{2}mv_A^2 + mgh = \frac{1}{2}mv_B^2 + mgh \Rightarrow v_B = v_A$$

$$\text{At point C: } \frac{1}{2}mv_A^2 + mgh = \frac{1}{2}mv_C^2 + mg\frac{h}{2} \Rightarrow v_C = \sqrt{v_A^2 + gh}$$

$$\text{At point D: } \frac{1}{2}mv_A^2 + mgh = \frac{1}{2}mv_D^2 \Rightarrow v_D = \sqrt{v_A^2 + 2gh}$$

7. Remember that potential energy has meaning only for a force that is conservative. A conservative force is that for which the work done in going from point A to point B is independent of the path chosen. Friction is an example of a non-conservative force and a potential energy cannot be defined. For a conservative force, $F = -\frac{dV}{dx}$. So, for

a spring, $V = \frac{1}{2}kx^2$ and so $F = -kx$.

8. Derivation of $F = -\frac{dV}{dx}$: If the particle moves distance Δx in a potential V , then change in PE is ΔV where, $\Delta V = -F \Delta x$. From this, $F = -\frac{\Delta V}{\Delta x}$. Now let $\Delta x \rightarrow 0$.

$$\text{Hence, } F = -\lim_{\Delta x \rightarrow 0} \frac{\Delta V}{\Delta x} = -\frac{dV}{dx}$$

Momentum depends on mass and velocity. $\vec{P} = m\vec{v}$
 (MLT^{-1})
 $\rightarrow kgms^{-1}$
 Ns

Summary of Lecture 9 - MOMENTUM

1. Momentum is the "quantity of motion" possessed by a body. More precisely, it is defined as:

Mass of the body $\times$ Velocity of the body. $P = mv$

The dimensions of momentum are MLT^{-1} and the units of momentum are $kg \cdot m/s$ or Ns

2. Momentum is a vector quantity and has both magnitude and direction, $p = m\vec{v}$. We can easily see that Newton's Second Law can be reexpressed in terms of momentum. When

I wrote it down originally, it was in the form $m\vec{a} = \vec{F}$. But since $\frac{d\vec{v}}{dt} = \vec{a}$, this can also be

written as $\frac{d\vec{p}}{dt} = \vec{F}$ (new form). In words, the rate of change of momentum of a body equals the total force acting upon it. Of course, the old and new are exactly the same, *and in the direction of that force.*

$$\frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = m \frac{d\vec{v}}{dt} = m\vec{a} = \vec{F}$$

Conservation of Linear Momentum:-

$$\sum \vec{F}_{ext} = 0 \rightarrow \frac{d\vec{P}}{dt} = 0$$

3. When there are many particles, then the total momentum $\vec{P}$ is:

Total vector sum

$$\vec{P} = \vec{p}_1 + \vec{p}_2 + \dots + \vec{p}_N$$

$$\frac{d\vec{P}}{dt} = \frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} + \dots + \frac{d\vec{p}_N}{dt}$$

$$= \vec{F}_1 + \vec{F}_2 + \dots + \vec{F}_N = \vec{F} \Rightarrow \sum_{i=1}^N \vec{F}_i = \text{Total external force}$$

$\vec{P}$ is conserved for isolated system. $\rightarrow \vec{P}$ constant

This shows that when there are several particles, the rate at which the total momentum changes is equal to the total force. It makes sense!

4. A very important conclusion of the above is that if the sum of the total external forces

vanishes, then the total momentum is conserved, $\sum \vec{F}_{ext} = 0 \Rightarrow \frac{d\vec{P}}{dt} = 0$. This is quite

independent of what sort of forces act between the bodies - electric, gravitational, etc. - or how complicated these are. We shall see why this is so important from the following examples.

$$F = -F' \Rightarrow F + F' = 0 \Rightarrow m_1 v_1' - m_1 v_1 + m_2 v_2 - m_2 v_2' = 0$$

$$m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$$

5. Two balls, which can only move along a straight line, collide with each other. The initial momentum is $P_i = m_1 u_1 + m_2 u_2$ and the final momentum is $P_f = m_1 v_1 + m_2 v_2$. Obviously one ball exerts a force on the other when they collide, so its momentum changes. But, from the fact that there is no external force acting on the balls, $P_i = P_f$, or $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$.

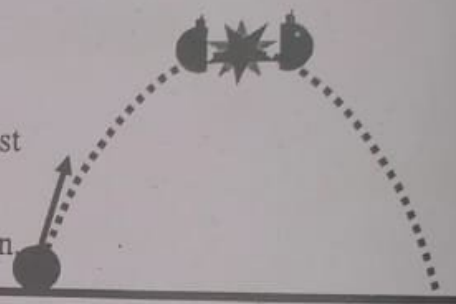
6. A bomb at rest explodes into two fragments. Before the explosion the total momentum is

zero. So obviously it is zero after the explosion as well, $P_f = 0$. During the time that the explosion happens, the forces acting upon the pieces are very complicated and changing rapidly with time. But when all is said and done, there are two pieces flying away with a total zero final momentum $P_f = m_1 v_1 + m_2 v_2$. Hence $m_1 v_1 = -m_2 v_2$. In other words, the fragments fly apart with equal momentum but in opposite directions. The centre-of-mass stays at rest. So, knowing the velocity of one fragment permits knowing the velocity of the other fragment.



7. If air resistance can be ignored, then we can do some interesting calculations with what we have learned.

So, suppose a shell is fired from a cannon with a speed 10 m/s at an angle 60° with the horizontal. At the highest point in its path it explodes into two pieces of equal masses. One of the pieces retraces its path to the cannon. Let us find the velocity of the other piece immediately after the explosion.



Solution: After the explosion: $P_{1x} = -5 \frac{M}{2}$ (why?). But $P_{1x} + P_{2x} = P_x = M \times 10 \cos 60$

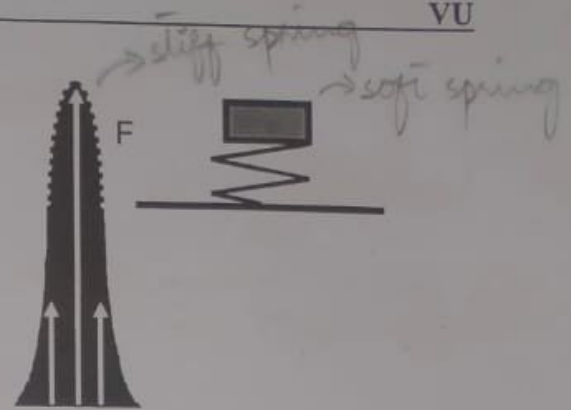
$$\Rightarrow P_{2x} = 5M + 5 \frac{M}{2}. \text{ Now use: } P_{2x} = \frac{M}{2} v_{2x} \Rightarrow v_{2x} = 15 \text{ m/s.}$$

8. When you hit your thumb with a hammer it hurts, doesn't it? Why? Because a large amount of momentum has been destroyed in a short amount of time. If you wrap your thumb with foam, it will hurt less. To understand this better, remember that **force is the rate of change of momentum: $F = \frac{dp}{dt} \Rightarrow dp = Fdt$. Now define the impulse I as: force $\times$ time over which the force acts.**

If the force changes with time between the limits, then one should define I as,

$I = \int_{t_1}^{t_2} F dt$. Since $\int_{t_1}^{t_2} F dt = \int_{p_i}^{p_f} dp$, therefore $I = p_f - p_i$. In words, **the change of momentum equals the impulse, which is equal to the area under the curve of force versus time.** Even if you wrap your thumb in foam, the impulse is the same. But the force is definitely not!

9. Sometimes we only know the force numerically (i.e. there is no expression like $F = \text{something}$). But we still know what the integral means: it is the area under the curve of force versus time. The curve here is that of a hammer striking a table. Before the hammer strikes, the force is zero, reaches a peak, and goes back to zero.



{ Impulse & Momentum }

$$F = \frac{dp}{dt} \Rightarrow dp = F dt$$

$$\text{Define: } I = \int_{t_1}^{t_2} F dt$$

$$\text{Since } \int_{t_1}^{t_2} F dt = \int_{p_i}^{p_f} dp \therefore I = p_f - p_i$$

Summary of Lecture 10 - COLLISIONS

1. Collisions are extremely important to understand because they happen all the time - electrons collide with atoms, a bat with a ball, cars with trucks, star galaxies with other galaxies, ... In every case, the sum of the initial momenta equals the sum of the final momenta. This follows directly from Newton's Second Law, as we have already seen.

2. Take the simplest collision: two bodies of mass m_1 and m_2 moving with velocities u_1 and u_2 . After the collision they are moving with velocities v_1 and v_2 . From momentum conservation

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \Rightarrow m_1 (u_1 - v_1) = m_2 (v_2 - u_2)$$

This is as far as we can go. There are two unknowns but only one equation. However, if the collision is elastic then,

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \Rightarrow \frac{1}{2} m_1 (u_1^2 - v_1^2) = \frac{1}{2} m_2 (v_2^2 - u_2^2)$$

Combine the two equations above,

$$u_1 + v_1 = v_2 + u_2 \Rightarrow u_1 - u_2 = v_2 - v_1$$

In words, this says that in an elastic collision the relative speed of the incoming particles equals the relative speed of the outgoing particles.

3. One can solve for v_1 and v_2 (please do it!) easily and find that:

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 + \left(\frac{2m_2}{m_1 + m_2} \right) u_2$$

$$v_2 = \left(\frac{2m_1}{m_1 + m_2} \right) u_1 + \left(\frac{m_2 - m_1}{m_1 + m_2} \right) u_2$$

Notice that if $m_1 = m_2$, then $v_1 = u_2$ and $v_2 = u_1$. So this says that after the collision, the bodies will just reverse their velocities and move on as before.

4. What if one of the bodies is much heavier than the other body, and the heavier body is at rest? In this case, $m_2 \gg m_1$ and $u_2 = 0$. We can immediately see that $v_1 = -u_1$ and $v_2 = 0$. This makes a lot of sense: the heavy body continues to stay at rest and the light body just bounces back with the same speed. In the lecture, you saw a demonstration of this!

5. And what if the lighter body (rickshaw) is at rest and is hit by the heavier body (truck)? In this case, $m_2 \ll m_1$ and $u_2 = 0$. From the above equation we see that $v_1 = u_1$ and $v_2 = 2u_1$. So the truck's speed is unaffected, but the poor rickshaw is thrust in the direction of the truck at twice the truck's speed!

6. Sometimes we wish to slow down particles by making them collide with other particles.

A system in which kinetic energy does not change in other form of energy and remain same is called elastic collision.

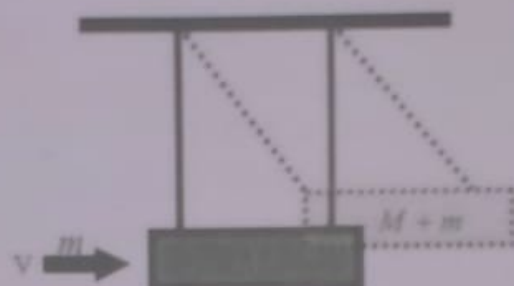
In a nuclear reactor, neutrons can be slowed down in this way. let's calculate the fraction by which the kinetic energy of a neutron of mass m_1 decreases in a head-on collision with an atomic nucleus of mass m_2 that is initially at rest.

Solution: $\frac{K_i - K_f}{K_i} = 1 - \frac{K_f}{K_i} = 1 - \frac{v_1^2}{v_i^2}$

For a target at rest: $v_2 = \left(\frac{m_2 - m_1}{m_2 + m_1} \right) v_i$

$\frac{K_i - K_f}{K_i} = \frac{4m_1 m_2}{(m_1 + m_2)^2}$

7. A bullet with mass m , is fired into a block of wood with mass M , suspended like a pendulum and makes a completely inelastic collision with it. After the impact, the block swings up to a maximum height y . What is the initial speed of the bullet?



Solution:

By conservation of momentum in the direction of the bullet,

$$mv = (m + M)V \Rightarrow v = \frac{(m + M)V}{m}$$

The block goes up by distance y , and so gains potential energy. Now we can use the conservation of energy to give, $\frac{1}{2}(m + M)V^2 = (m + M)gy$, where V is the velocity acquired by the block+bullet in the upward direction just after the bullet strikes. Now use $V = \sqrt{2gy}$. So finally, the speed of the bullet is: $v = \frac{(m + M)}{m} \sqrt{2gy}$.

8. In 2 or 3 dimensions, you must apply conservation of momentum in each direction separately. The equation $\vec{P}_i = \vec{P}_f$ looks as if it is one equation, but it is actually 3 separate equations: $P_{ix} = P_{fx}$, $P_{iy} = P_{fy}$, $P_{iz} = P_{fz}$. On the other hand, suppose you had an elastic collision. In that case you would have only one extra equation coming from energy conservation, not three.
9. What happens to energy in an inelastic collision? Let's say that one body smashes into another body and breaks it into 20 pieces. To create 20 pieces requires doing work against the intermolecular forces, and the initial kinetic energy is used up for this.